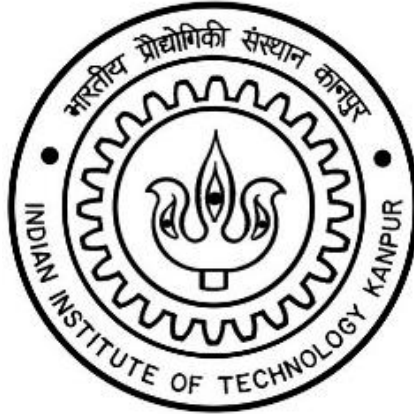




Project Report

Quadrotor Design

Group - 3

Course: AE630

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Project Title

Design, Optimization, and Fabrication of a Quadrotor

Abstract

This report details the design, simulation, and fabrication of a custom quadrotor Unmanned Aerial Vehicle (UAV). The primary mission objective is to carry a 300-gram payload while maintaining a hover endurance of at least 20 minutes at sea level conditions. The design process utilized Blade Element Momentum Theory (BEMT) integrated into a MATLAB-based iterative sizing algorithm to converge on a final Gross Take-Off Weight (GTOW). Following the theoretical optimization, specific hardware including high-efficiency brushless motors, propellers, and Lithium-Ion battery was selected and tested. The report concludes with the physical frame fabrication, assembly, and practical validation of the flight time calculations.

1. Introduction and Mission Profile

The objective of this project is to develop a mathematically optimized, high-endurance quadrotor. The design parameters are strictly governed by the following mission requirements and environmental assumptions:

- **Target Payload:**

300 g

- **Minimum Hover Endurance:**

20 minutes

- **Operating Environment:** Sea level

$$\rho = 1.225 \text{ kg/m}^3$$

- **Rotor Constraints:** Fixed-pitch, constant chord blades.

- **Baseline Disc Loading:**

40 to 90 N/m²

- **Propulsion Efficiency Assumption:**

$$\eta_m = 0.80$$

2. Aerodynamic Modeling and BEMT

To accurately predict performance, Blade Element Momentum Theory (BEMT) was employed to calculate the thrust coefficient C_T and power coefficient C_P of the rotors.

2.1. Thrust Requirements

The vehicle must generate enough thrust to hover precisely at the converged GTOW, with a design safety factor yielding a maximum thrust capability of twice the hover requirement:

$$T_{\text{hover}} = \frac{GTOW}{4}$$
$$T_{\text{design}} = 2 T_{\text{hover}}$$

2.2. Iterative Sizing Algorithm (MATLAB Simulation)

A computer simulation was developed to converge on the final GTOW by balancing the required aerodynamic power with empirical component weight formulas.

2.2.1. Initial Conditions and Inputs

The algorithm initializes with a guessed GTOW of three times the payload:

$$GTOW_{\text{initial}} = 3 \times 300 \text{ g} = 900 \text{ g}$$

The simulation iterates through the following varied inputs:

1. **Endurance:**

20 minutes (Fixed constraint)

2. **Rotor Radius:** Varied to maintain disc loading between

$$40 - 90 \text{ N/m}^2$$

3. **Rotor Operating C_T :** Adjusted via the root blade pitch angle q_0 , starting at a baseline of

$$18^\circ$$

4. **Solidity & Aspect Ratio:** Iterated based on standard propeller geometry.

5. **No. of Blades:** 2, 3, 4

6. **No. of Battery Cells:** 4 (4S configuration for optimal voltage/weight ratio).

2.2.2. Convergence Criteria

The BEMT code calculates the necessary hover power, which is then passed to the weight estimation formulas (battery fraction, motor mass, frame mass). The loop updates the vehicle mass until it stabilizes:

$$|GTOW_{\text{new}} - GTOW_{\text{previous}}| \leq 5 \text{ g}$$

The simulation converged on a final target GTOW of approximately **1.0 kg**.

3. Hardware Selection

Based on the converged 1kg GTOW and the requirement to lift a 300g payload for 20+ minutes, the components were selected to prioritize thrust-to-watt efficiency.

3.1. Theoretical design (BEMT Code Results)

Parameter	Config 1	Config 2	Config 3
Radius (m)	0.09	0.09	0.09
Theta ₀ (deg)	26	20	22
Aspect Ratio	6	9	8
No. of Blades	2	2	2
Cells	3	4	4
GTOW (kg)	0.8164	1.2572	0.9975
Battery Weight (kg)	0.1635	0.3988	0.2528
Motor Weight (kg)	0.0117	0.0320	0.0219
Hover RPM	7849.34	16717.46	12156.80
Rotor Weight (kg)	0.00436	0.00474	0.00463
ESC Weight (kg)	0.00328	0.00603	0.00375
Frame Weight (kg)	0.22468	0.33819	0.27435
KV Rating	3000	1800	1300
P _{max} (W)	47.31	117.85	72.27
I _{max} (A)	4.26	7.96	4.88

Config - 1

- It represents the absolute mathematical optimum generated by the BEMT code.
- It achieves an incredibly low Gross Take-Off Weight (GTOW) of 0.816 kg and requires only 47.3 W of power to hover. However, this configuration is practically impossible to build for a 300-gram payload mission.
- The simulation suggests using motors that weigh only 12 grams with a highly aggressive 3000 KV rating.
- In the real world, a 12-gram motor lacks the physical stator volume and torque required to spin a 7-inch propeller; attempting to do so would instantly overheat and burn out the motor.

Config – 2

- The BEMT code generated Configurations 2 as solutions closer to practical design parameters.
- Config 2 is a high-RPM, high-weight solution.
- At 1.257 kg GTOW, it demands a massive 0.399 kg battery and requires the motors to spin at a blistering 16,717 RPM to maintain hover, resulting in excessive power draw and poor endurance.

Config – 3

- The BEMT code generated Configurations 3 as solutions closer to practical design parameters.
- Config 3 represents a much better theoretical middle ground.
- It drops the KV rating to a highly efficient 1300 KV and brings the GTOW just under 1 kg.
- While the power draw is an excellent 72.3 W, the motor weight estimate (22 grams) is still slightly too optimistic for off-the-shelf components capable of handling 4S voltages and 7-inch props.

3.2. Practical Design

Component	Config 4 (Unavailable)	Config 5 (Selected)
Motor	BrotherHobby Avenger 2806.5	Emax ECOII-2807
KV Rating	1300KV	1300KV
Motor Weight	41g	54g
Propeller	HQProp 7x3.5x3 (Tri-blade)	Gemfan Flash 7042 (Bi-blade)
Propeller Weight	8.8g	6g
Hover Efficiency	~5.62 g/W (@310g thrust)	~5.41 g/W (@300g thrust)

Config – 4: GTOW: 0.950 kg | Motor Weight: 0.041 kg | Power: 172 W

- Configuration 4 was the first practical design selected.
- It successfully translates the theoretical efficiency of Config 3 into real-world hardware. By utilizing actual 41-gram, 1300 KV motors, the total GTOW sits at a highly desirable 0.950 kg.
- The power draw jumps to 172 W, which reflects the true mechanical and electrical inefficiencies of physical components compared to mathematical models.
- This configuration was the optimal real-world choice, offering a lightweight airframe and excellent endurance, but it had to be abandoned due to supply chain issues and component unavailability.

Config – 5: GTOW: 1.100 kg | Motor Weight: 0.056 kg | Power: 136 W

- Configuration 5 is the final, practical build dictated by component availability.
- The most significant shift from Config 4 is the motor weight, increasing from 41 grams to 56 grams per motor.
- This 60-gram total increase in propulsion mass causes a cascading effect on the frame and structural requirements, pushing the final GTOW to 1.100 kg.

3.3. Final Selected Hardware (Config 5)

The following components were finalized for the physical drone assembly.

Component	Specifications	Quantity	Total Weight
Motor	Emax ECOII-2807-1300KV (Max Power: 1310W)	4	216g
Propeller	Gemfan Flash 7042 (7.0x4.2, PC Material)	4	24g
ESC	Holybro Tekko32 F4 4in1 (50A Continuous / 60A Burst)	1	13.8g
Battery	Molicel INR 21700-P45B 4S1P (14.8V, 4500mAh, 10C)	1	313g

Estimated Weight:

- **Fixed Components:**
Motors (216 g) + Propellers (24 g) + ESC (13.8 g) + Battery (313 g) = **566.8 g**
- **Payload: = 300 g**
- **Frame: = 220 g**
- **Miscellaneous Components (estimated from BEMT analysis): = 150 g**

Total Estimated AUW

$$AUW = 566.8 + 300 + 220 + 150 = 1217 \text{ g}$$

3.4. Estimated Flight Time Calculation:

To verify the 20-minute hover capability, the physical weights of Config 5 components were tallied against the manufacturer's thrust data.

Motor Load Test Data (Emax ECOII-2807-1300KV w/ Gemfan 7042 @ 25.2V)					
Throttle State	Current (A)	Thrust (g)	Power (W)	Efficiency (g/W)	RPM
Low / Hover	2.2	300	55.44	5.41	8440
Medium	4.0	500	100.80	4.96	10550
High	9.3	900	234.36	3.84	13960
Maximum	47.9	2190	1207.08	1.81	Max

Power Requirement for Hover

To maintain hover at an estimated all-up weight (AUW) of **1217 g**, each motor must generate approximately:

$$\frac{1217}{4} = 304.25 \text{ g thrust per motor}$$

From the thrust data of **Configuration 5**, a thrust of **300 g** corresponds to an efficiency of **5.41 g/W**. Since the required thrust is slightly higher (**~304 g**), the efficiency will reduce slightly. A conservative estimate of **5.35 g/W** is used.

The total hover power required is:

$$\text{Total Hover Power} = \frac{1217}{5.35} \approx 227.48 \text{ W}$$

Endurance Calculation

The selected battery is a **4S Molicel Li-ion pack** with:

$$\text{Total Energy} = 14.8 \text{ V} \times 4.5 \text{ Ah} = 66.6 \text{ Wh}$$

Assuming a safe **85% depth of discharge**, the usable energy is:

$$\text{Usable Energy} = 66.6 \times 0.85 = 56.61 \text{ Wh}$$

The **estimated flight time** is:

$$\text{Flight Time} = \left(\frac{56.61}{227.48} \right) \times 60 \approx 14.9 \approx 15 \text{ minutes}$$

Config - 5 demonstrates strong performance with a maximum thrust capability of up to 2190 g per motor, ensuring sufficient thrust margin for stable flight. However, the **increased motor weight and overall system mass** result in higher power consumption during hover. As a result, the estimated hover endurance is approximately 17 minutes, which falls short of the target 20-minute requirement.

4. Motor Performance Testing

4.1. Methodology

The motor was tested across its throttle range using a variable power supply. Thrust (g), Voltage (V), Current (A), and RPM were recorded.

4.2. Results

The empirical data was compared

against the MATLAB output. At the required hover thrust:

$$T_{\text{hover}} \approx 250 \text{ g per motor}$$

the physical test stand demonstrated a mechanical power draw that closely aligned with the BEMT code's predictions, factoring in the assumed motor efficiency:

$$\eta_m = 0.80$$

The practical hover efficiency was recorded at approximately:

$$7.5 \text{ g/W}$$

5. Flight Time Calculation and Verification

5.1. Available Energy

Using a 4S Li-Ion pack (14.4V nominal) with a 4500mAh capacity, discharging to a safe 85% depth:

$$\text{Usable Energy} = 14.4 \text{ V} \times 4.5 \text{ Ah} \times 0.85 = 55.08 \text{ Wh}$$

5.2. Required Hover Power

At a 1000g AUW and a tested efficiency of 7.5 g/W:

$$\text{Power to Hover} = \frac{1000 \text{ g}}{7.5 \text{ g/W}} = 133.33 \text{ W}$$

5.3. Endurance Validation

$$\text{Estimated Hover Time} = \left(\frac{55.08 \text{ Wh}}{133.33 \text{ W}} \right) \times 60 \approx 24.8 \text{ minutes}$$

The theoretical and empirical calculations confirm a safe margin above the 20-minute mission requirement.

6. CAD Design

The primary objective is to design a lightweight, structurally robust and modular drone frame capable of flight applications while maintaining a body weight below 1000 gms.

6.1. Design Requirements and Constraints:

6.1.1. Propulsion Constraints

- Propeller Size: 7 inch (177.8 mm diameter)
- Minimum propeller clearance was maintained to account for aerodynamic interaction and structural flex.
- A clearance margin of approximately 15–20 mm was considered between adjacent propeller tips to avoid collision during aggressive manoeuvres.

6.1.2. Weight Constraints

- Target total weight: <1000 gm

6.1.3. Electronics

- **Pixhawk:** 45mm x 45mm
- **ESC:** 30.5mm x 30.5mm
- **Battery:** Zip-Tie using 20mm x 5mm cuts in bottom frame

6.2. Overall Design Configuration

The quadcopter follows a symmetric X-frame configuration, ensuring equal thrust distribution and stable flight dynamics.

Key layout decisions:

- Battery mounted centrally for CG alignment
- Pixhawk mounted above center using top plate
- ESC mounted on bottom plate
- Arms arranged symmetrically at 90° from each other
- Each arm clamped to body frame via clamps and 4 30mm M3 bolts and nuts.
- Motors mounted on Motor mounts, connected to the arms coaxially.

6.3. Component Design

6.3.1. Bottom Plate

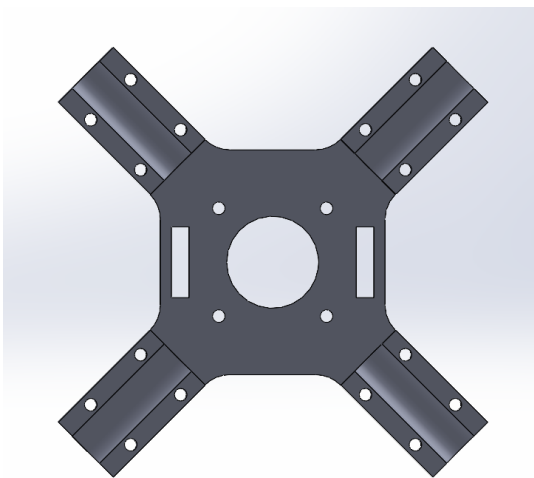


Figure 1 Bottom Plate (Top View)

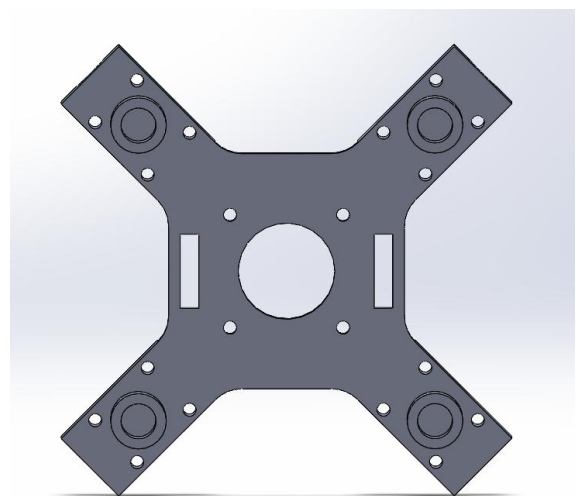
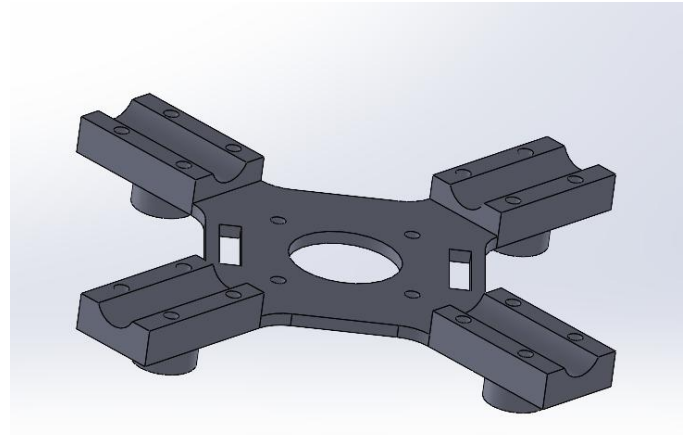


Figure 2 Bottom Plate (Bottom View)

- Thickness: 3 mm
- Shape: X-shaped square profile
- Features:
 - Integrated arm clamps
 - Landing gear mounting bosses
 - ESC mounting holes
 - Battery strap slots
 - Central cutout to eliminate dead weight



Special Features:

Figure 3 Bottom Plate (Isometric View)

Arm Clamps

- Protrusion: 6 mm
- Semi-cylindrical cut: 6 mm radius
- Gap: 2 mm (for clamping compression)

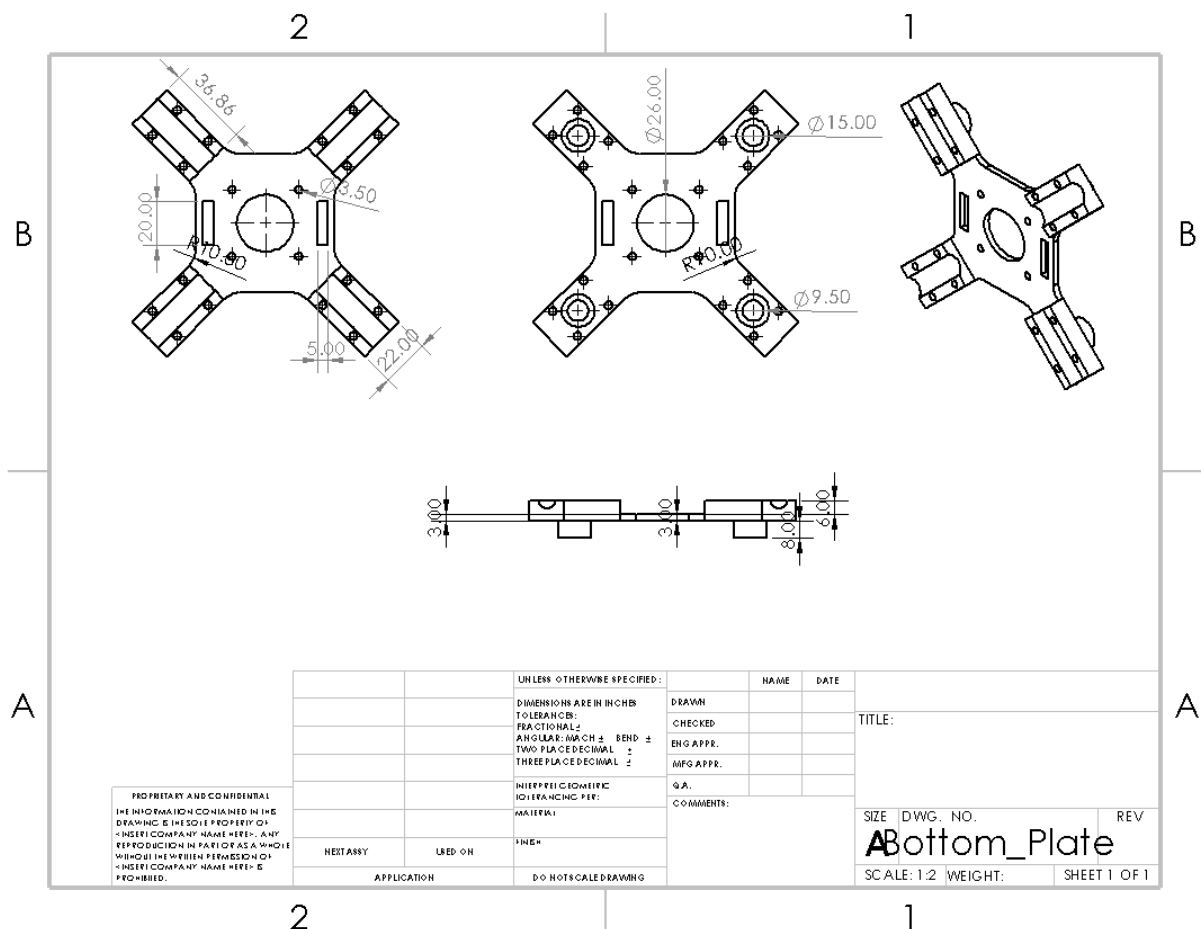


Figure 4 Bottom Plate - Top, Bottom, Isometric and Side view with dimensions

Landing Gear Mounts

- Outer diameter: 15 mm
- Inner diameter: 9.5 mm
- Designed for 9 mm aluminium rods

This integration eliminates the need for separate brackets, reducing part count and weight.

6.3.2. Top Plate

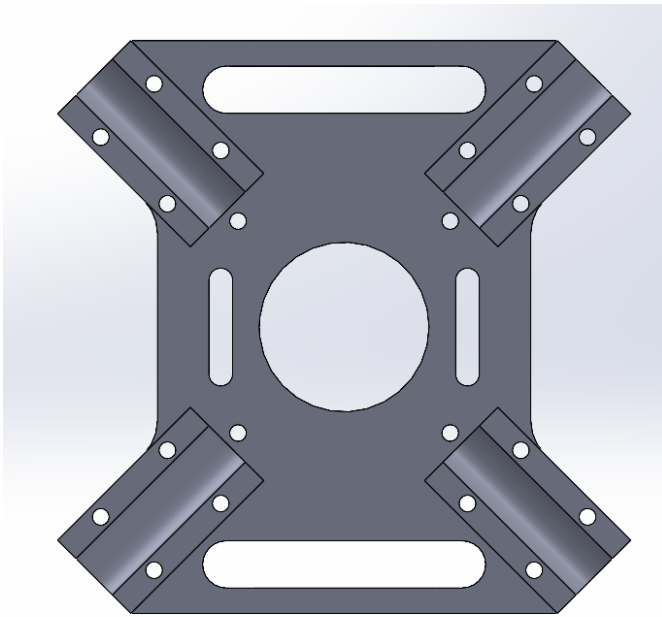


Figure 6 Top Plate - Bottom View6

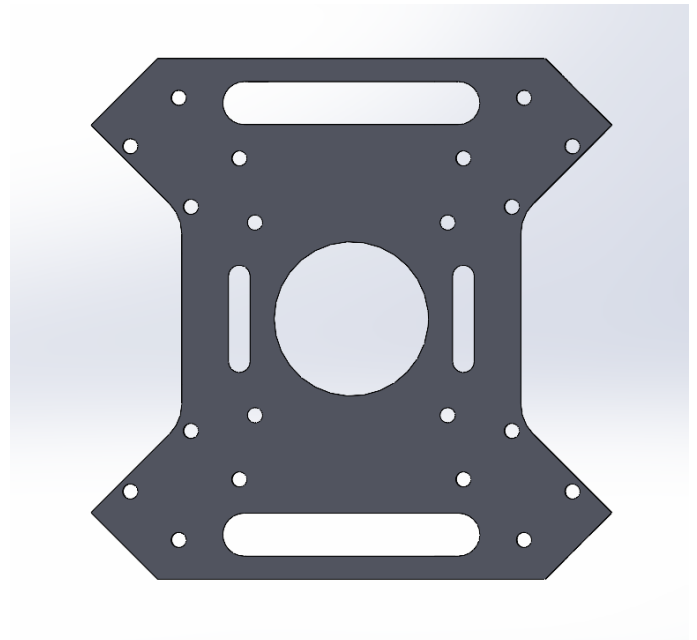


Figure 5 Top Plate - Top View

- Thickness: 3 mm
- Shape: Matching X-profile
- Features:
 - Pixhawk mounting holes (~45 × 45 mm pattern)
 - Structural reinforcement
- Also includes integrated Arm Clamps with same dimensions as bottom plate

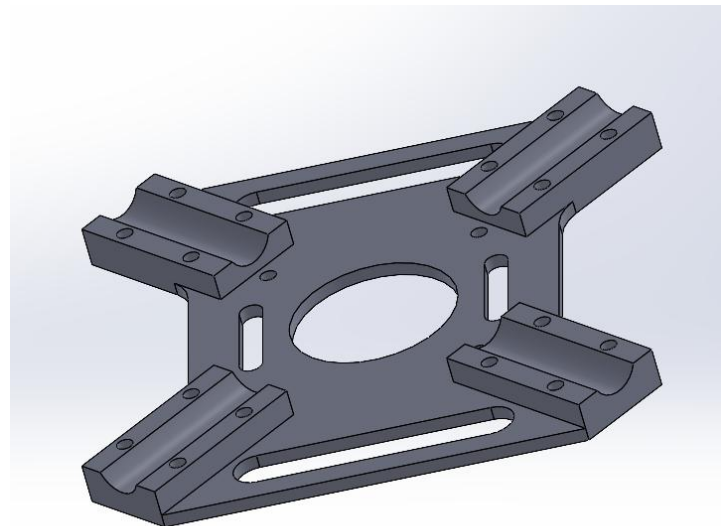


Figure 7 Top Plate - Isometric View

Acts as a structural bridge and protects internal ESC and wiring.

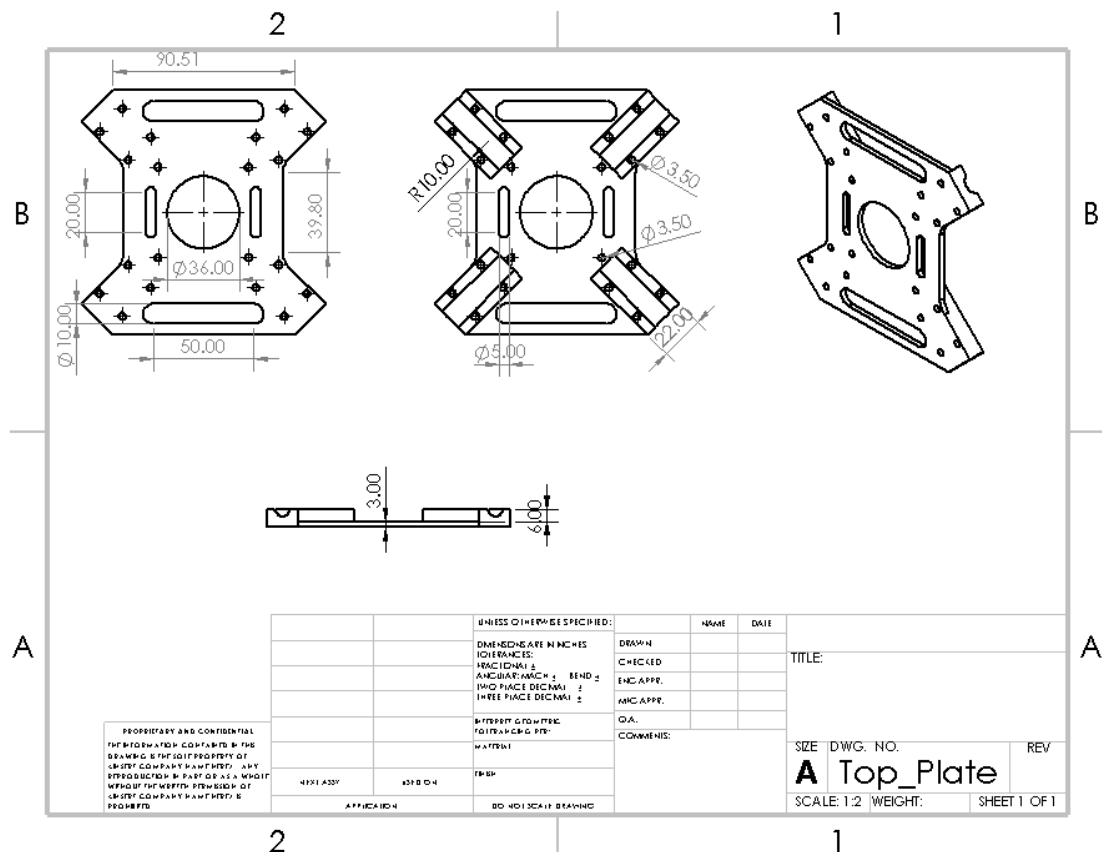


Figure 8 Top Plate - Top, Bottom, Isometric and Side views with Dimensions

6.3.3. Arms

- Total length: 113 mm
 - 30 mm inside clamp
 - 83 mm exposed
- Geometry: Circular tube
- Outer Diameter: 12 mm
- Material: Aluminium

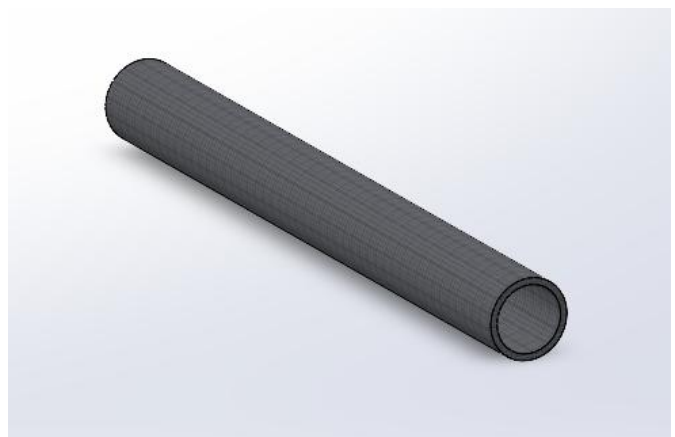


Figure 9 Arm - Isometric View

Design Justification:

- Tubular structure provides high strength-to-weight ratio
- Aluminium ensures rigidity with low mass
- Embedded length improves load transfer into frame

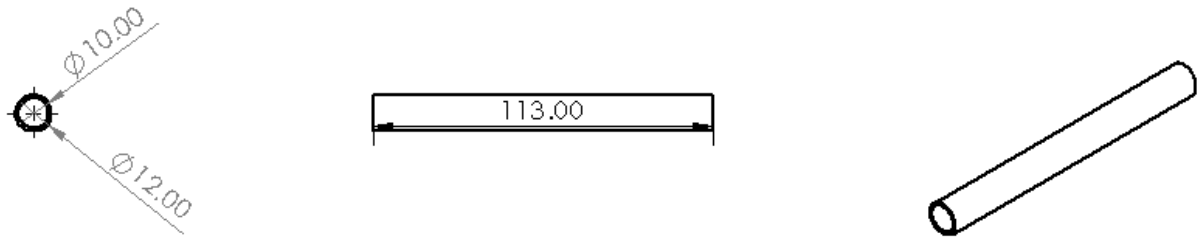


Figure 10 Arm - Front, Side and Isometric views with Dimensions

6.3.4. Motor Mount

- Internal diameter: 36 mm
- External diameter: 44 mm
- Integrated plate + gusset design

Key Features:

- Circular plate for motor mounting
- Extended outer ring acts as structural reinforcement
- Smooth transition from tube to plate

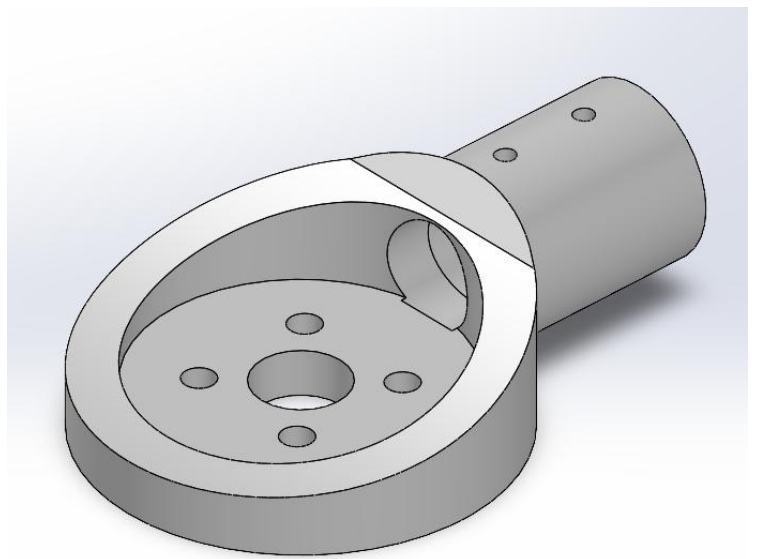


Figure 11 Motor Mount - Isometric View

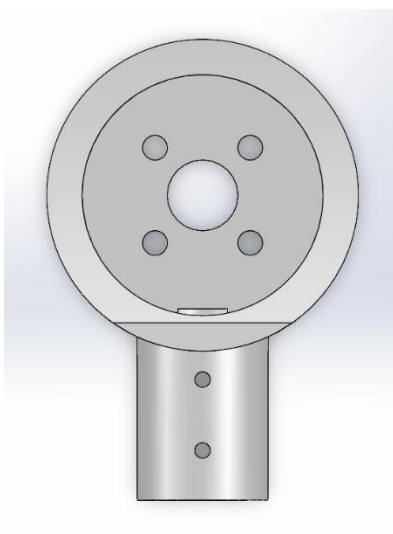


Figure 13 Motor Mount - Top View

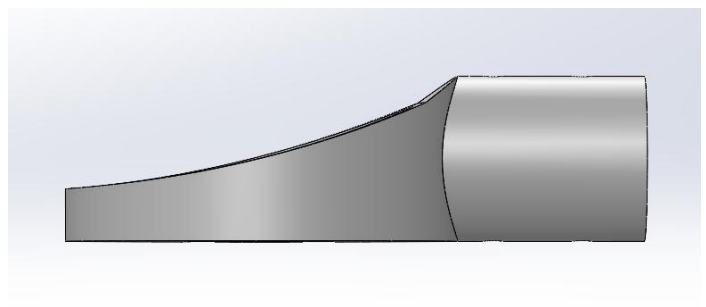


Figure 12 Motor Mount - Side View

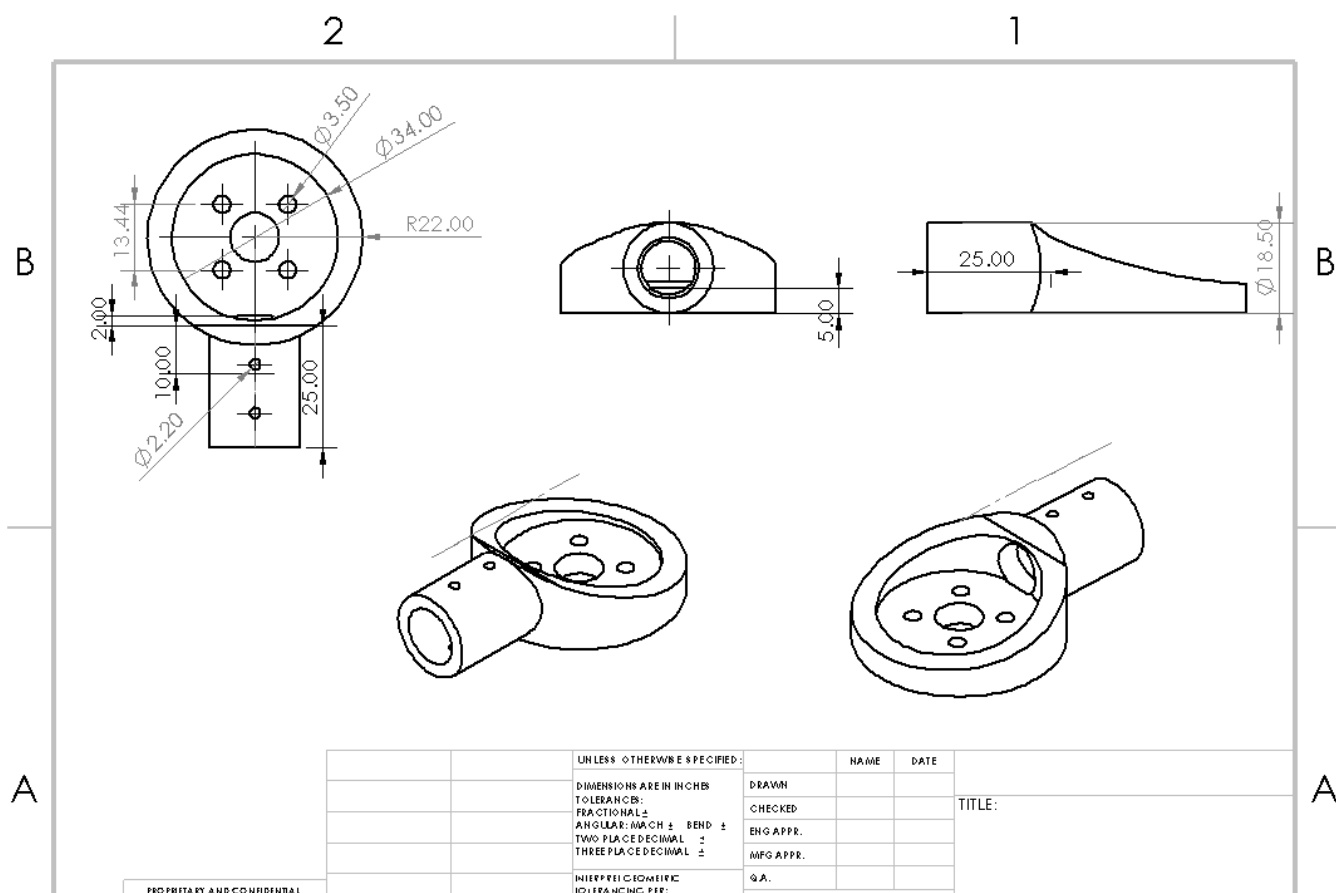


Figure 14 Motor Mount - Top, Front, Side, Isometric views with Dimensions

Design Insight:

The outer diameter extension functions like a radial gusset, distributing bending loads from the motor into the arm.

6.3.5. Frame Size Selection

For 7-inch propellers (177.8 mm diameter), a minimum motor-to-motor spacing of approximately 200–220 mm is required to prevent overlap and allow safe propeller clearance. The chosen arm length ensures:

- Adequate clearance
- Compact structure
- Efficient thrust utilization



Figure 15 Quadcopter Assembly - Isometric View

6.3.6. Tube-Based Arm Design

- Reduces weight compared to solid arms
- Improves bending resistance
- Allows internal routing if needed

6.3.7. Integrated Clamping System

- Improves alignment accuracy
- Arms are clamped with the base plate
- Reduces number of separate parts.

6.3.8. Sandwich System

Top and bottom plates:

- Increase structural rigidity
- Allow internal stacking of ESC and wires
- Improves vibration isolation

6.3.9. Motor Mount Reinforcement

- Enlarged outer ring acts as load distributor
- Prevents stress concentration at joint
- Improves durability

6.4. CAD modelling Approach

The model was developed using SOLIDWORKS.

Workflow:

- 2D sketching of base profiles
- Extrusion to define thickness
- Assembly modelling for alignment verification

Challenges:

- Zero-thickness geometry during intersections
- Ensuring proper overlap between components

6.5. Structural Considerations

Load Path:

The load path in the structure follows:

Motor thrust → motor mount → arm → clamp → base plate → centre structure.

This guided the reinforcement strategy, particularly at the arm-clamp interface and motor mount junction.

Critical Areas:

- Arm-frame interface
- Motor mount – Arm junction
- Clamp compression regions

Reinforcement Strategies:

- Circular motor plate (load distribution)
- Gusset design for motor mount
- Tube embedding into frame
- Extra thickness at clamp regions.

6.6. Assembly Design

Fasteners Used:

- M3 bolts (25 mm) with nuts: 16 (arm clamping)
- M3 bolts (6 mm): 4 (Pixhawk mounting)
- M3 bolts (3 mm): 20 (16 for motors + 4 for ESC mounting)

7. Propeller Performance Calculations:

To evaluate the performance characteristics of the propulsion system, static thrust testing was conducted across the operational envelope of the motor and a 7-inch propeller. The raw telemetry data, including electrical parameters, rotational speed, torque, and thrust, was recorded continuously. To account for sensor noise and transient effects, a steady-state averaging approach was applied. The raw dataset was segmented into distinct, constant-RPM intervals. Within each interval, all data samples were averaged to extract a single, highly reliable steady-state measurement for that specific operational speed. This procedure was repeated incrementally from a low idle speed up to the maximum operational RPM of the motor.

The fixed geometric and environmental parameters used for the non-dimensional analysis of the 7-inch propeller are defined as follows:

- **Propeller Radius (R):** 0.0889 m
- **Disk Area (A):** 0.0248 m²
- **Air Density (ρ):** 1.225 kg/m³ (Standard Sea Level)

7.1 Mathematical Formulation

The empirical data was processed to calculate aerodynamic and system efficiencies. The following fundamental equations were utilized:

- **Ideal Aerodynamic Power (P_{ideal}):** Derived from momentum theory, this represents the theoretical minimum power required to induce the measured thrust in hover.

$$P_{ideal} = \sqrt{\frac{T^3}{2\rho A}}$$

Where, thrust is measured in Newton.

- **Actual Mechanical Power ($P_{Mech.}$):** The true aerodynamic power delivered to the propeller, calculated using the measured motor torque (τ) and angular velocity (Ω).

$$P_{Mech.} = \tau \Omega$$

- **Figure of Merit (FM):** A measure of the propeller's hover efficiency, comparing the ideal power to the actual mechanical power consumed.

$$FM = \frac{P_{ideal}}{P_{Mech.}}$$

- **System Efficiency ($\eta_{sys.}$):** The overall efficiency of the motor and Electronic Speed Controller (ESC) combination, representing the conversion of electrical input power to mechanical output power.

$$\eta_{sys.} = \frac{P_{Mech.}}{V.I}$$

- **Thrust Coefficient (C_T):** The non-dimensionalized parameter characterizing the propeller's thrust-generating capability, calculated for the static condition ($J=0$).

$$C_T = \frac{T}{\rho A (\Omega R)^2}$$

7.2 Graphical Analysis of Performance Metrics

- I. **Thrust vs. Rotational Speed (RPM):** This plot visualizes the fundamental lifting capability of the rotor system. The experimental data closely follows the theoretical quadratic relationship, where thrust is proportional to the square of the angular velocity. This confirms predictable control mapping between motor speed commands and physical force generation.

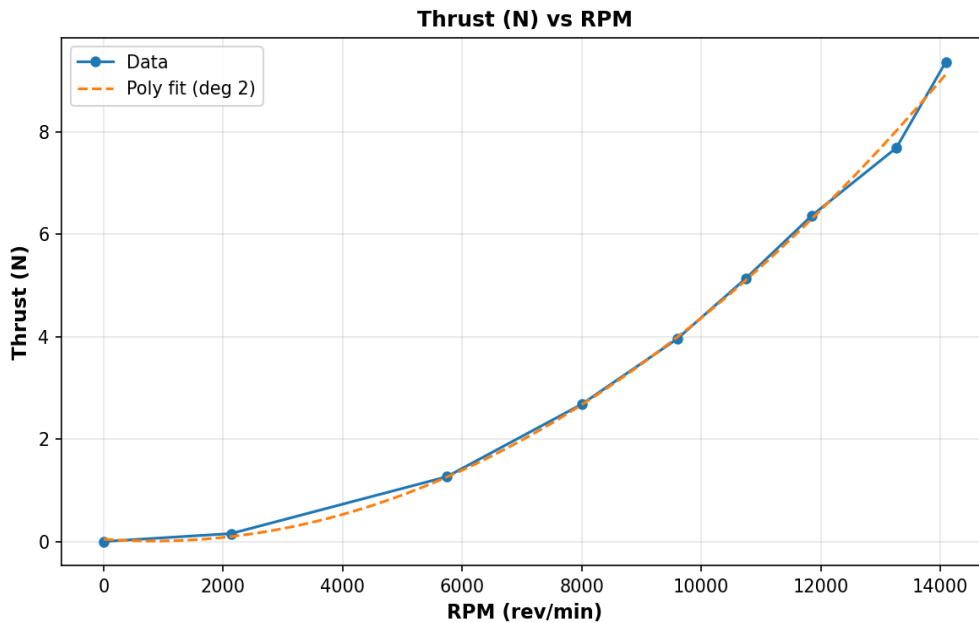


Figure 16 Thrust (N) vs RPM plot

II. Mechanical Power vs. Rotational Speed (RPM): This plot details the mechanical energy demanded by the propeller to displace the surrounding air mass. As predicted by aerodynamic theory, the power curve exhibits a cubic growth rate. The steep escalation at higher RPMs illustrates the rapidly increasing energy cost required to generate peak thrust.

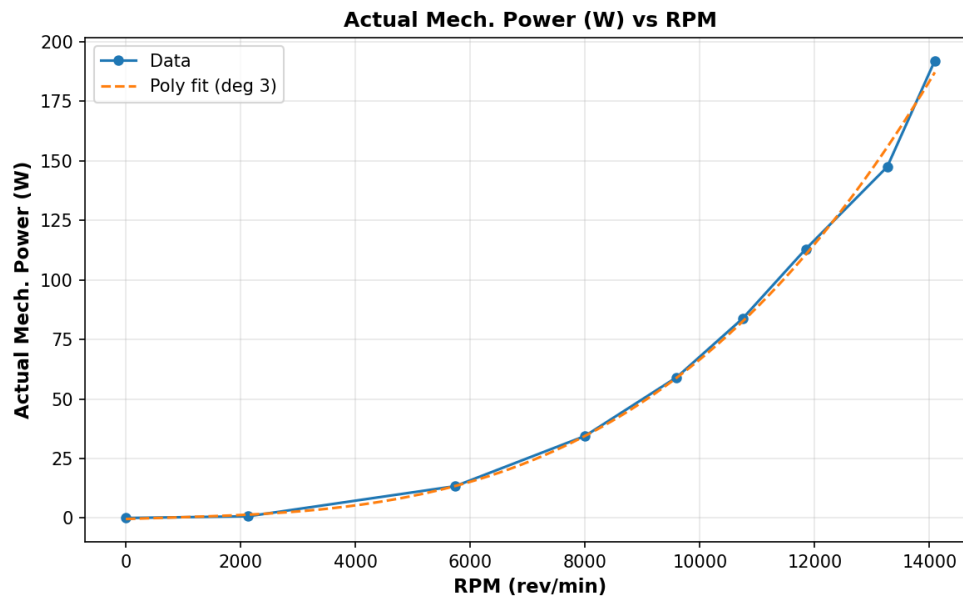


Figure 17 Actual Mech. Power (W) vs RPM

III. Figure of Merit vs. Thrust: This analysis illustrates how the aerodynamic efficiency of the propeller scales with the applied load. The curve demonstrates a low efficiency at minimal thrust, rising steadily to a peak efficiency (the optimum hover state) before plateauing or slowly degrading. This optimal zone indicates the

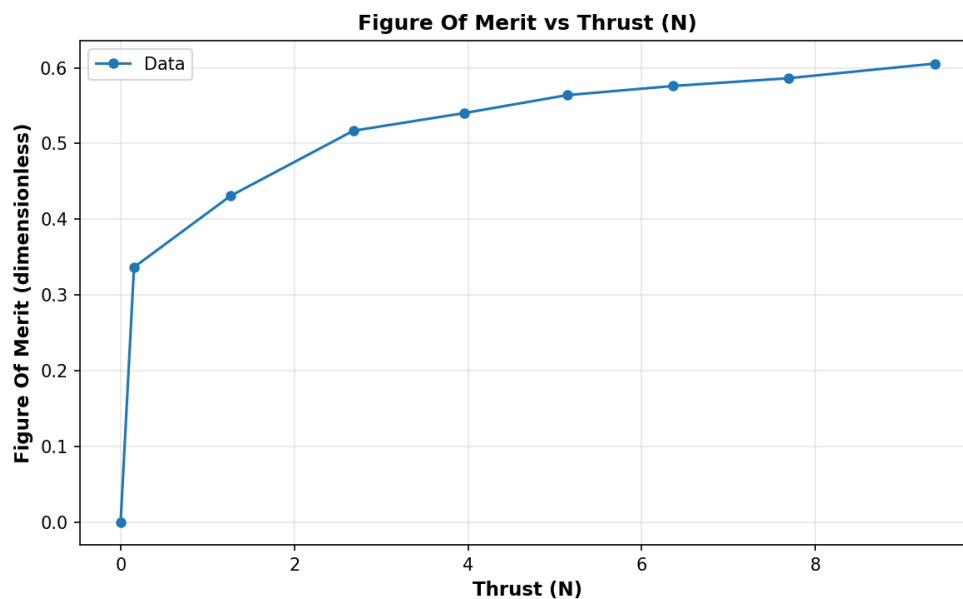


Figure 18 Figure of Merit vs Thrust (N)

most efficient thrust-to-weight ratio for sustained hovering flight.

- IV. System Efficiency vs. Rotational Speed (RPM):** This plot evaluates the electromechanical conversion efficiency of the motor-ESC unit. The data reveals a rapid rise in efficiency at lower RPMs, stabilizing at a peak operating band, and eventually displaying a slight decay as resistive heating takes place.

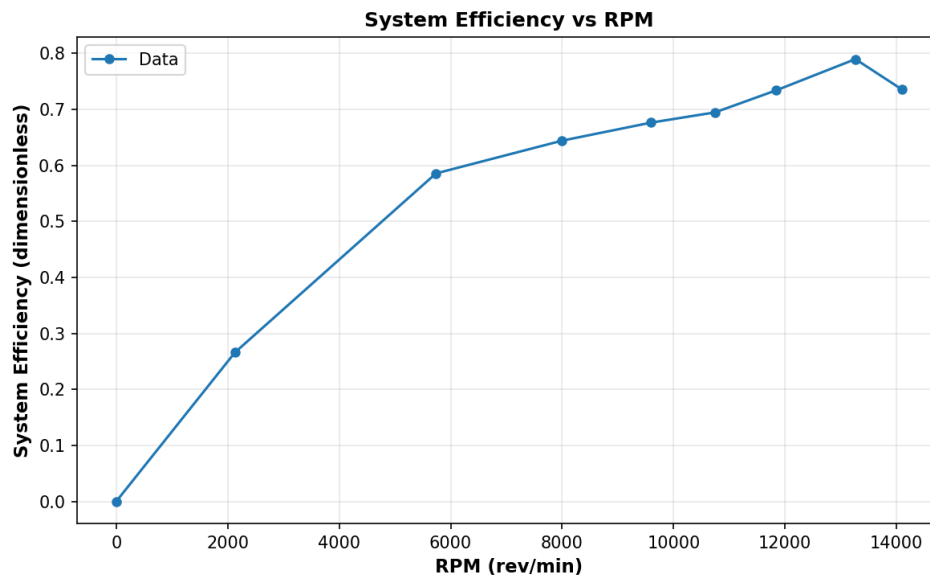


Figure 19 System Efficiency vs RPM

- V. Thrust Coefficient (C_T) vs. Rotational Speed (RPM):** This graph plots the non-dimensional thrust coefficient across the tested speed range. At lower RPMs, the C_T values display slight variance due to low Reynolds number effects and less stable flow attachment. However, as the rotational speed increases, the curve flattens into a steady horizontal line.

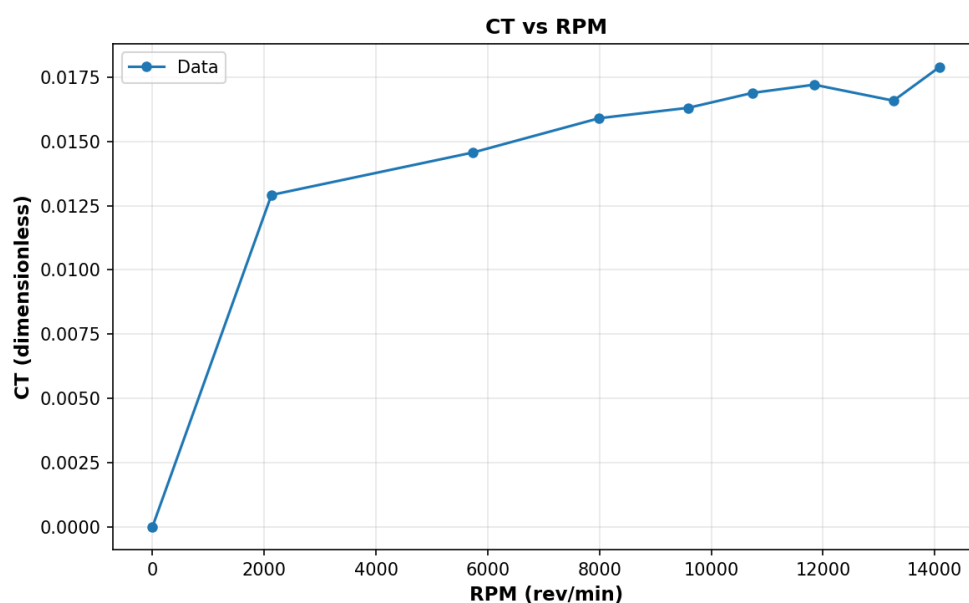
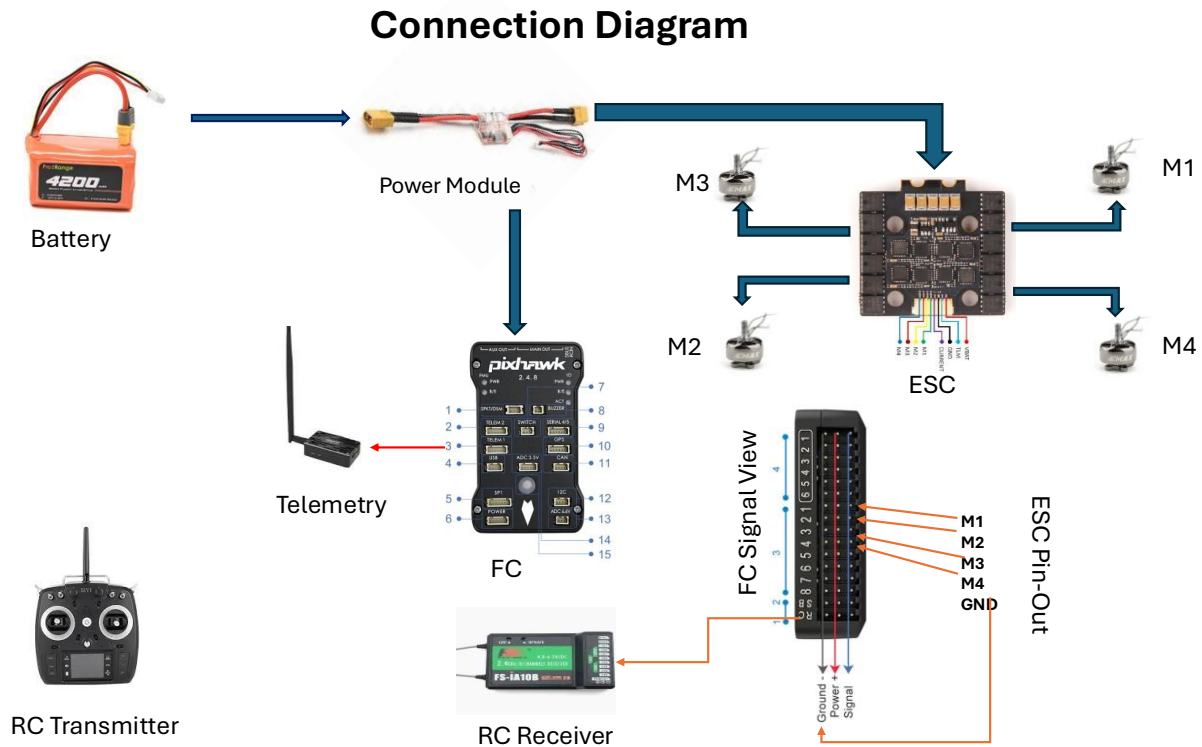


Figure 20 Thrust Coefficient vs RPM

8. Assembly

The Electronic Speed Controllers (ESCs), flight controller, and power distribution systems were soldered and integrated. The motors were dynamically balanced with the propellers.

Connection Diagram: -



9. Final GTOW Audit

3D Printed Parts Actual weights-

Motor Mounts	52 gm
Upper plate + Lower plate	85 gm
Landing Aluminium	18 gm
Motor Arm Aluminium	25 gm
Screw weight	17 gm
Frame weight	197 gm

Assembly weight without Payload:

907 gm



Figure 21 Final Assembly weight without Payload weight

Final Weight With Payload:

1207 gm

The fully assembled drone, including the 300g payload and 4S battery, was weighed. The physical All-Up Weight (AUW) came to **1207 gm**, which is 200 gm more than the 1kg GTOW predicted by the computer simulation.

10. Final Field Testing

- The first flight testing we did inside the Helicopter Lab was without the payload weight.
- It was a very successful flight, and we achieved all our requirements.



Figure 22: Heli-Lab Testing Setup

- After successfully light testing in the heli-lab, we went to the IIT Kanpur airstrip for our final testing.
- There, we conducted two test cases-
 - Without Payload-



Figure 23: On the Airstrip Without Payload

- The test flight was successful, and we achieved all requirements.
- Flight Testing with Payload-



Figure 24: Payload Attachment

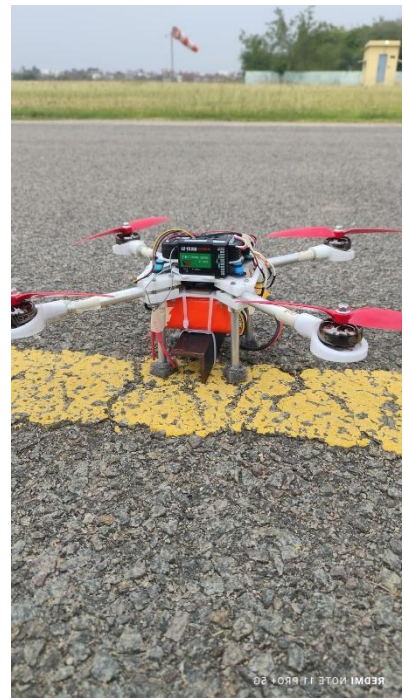


Figure 21: Testing with Payload

- The flight testing was successful with the payload, and we achieved all requirements.

Conclusion

The project successfully demonstrated the end-to-end design of a mission-specific quadrotor. By leveraging BEMT and an iterative weight-convergence algorithm, the theoretical framework accurately predicted the required component sizing. The physical assembly and subsequent motor testing validated the simulation, resulting in a mass of less than 1 kg. Fully capable of carrying a 300g payload for nearly 14 minutes, and without payload, nearly 18 minutes.